Math 520, F07 **Name:** Answers
Quiz 6 **SSN:** xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. You must simplify your answer if possible.

Problem 1. (20 points) Solve the following initial-value problem

$$2y'' - 6y' + 4y = te^{3t} + 1; \quad y(0) = 1, \ y'(0) = 5/8.$$

Solution: step 1. Find two linearly independent solutions of the homogeneous equation $2y'' - 6y' + 4 = 0$

$$2r^2 - 6r + 4 = 0$$

$$(2r-2)(r-2) = 0 \Rightarrow r_1 = 1,\ r_2 = 2$$

$$\Rightarrow y_1(t) = e^t, \quad y_2(t) = e^{2t}$$

$$W[y_1, y_2](t) = e^t \cdot (2e^{2t}) - e^t e^{2t} = e^{3t}$$

step 2. Find a particular solution ψ for $2y'' - 6y' + 4 = te^{3t} + 1$.

$$y'' - 3y' + 2 = \underbrace{\frac{te^{3t}+1}{2}}_{=g(t)} \text{ (normalization)}$$

$$\psi(t) = u_1(t)y_1(t) + u_2(t)y_2(t)$$

$$u_1'(t) = \frac{-g(t)y_2(t)}{W[y_1,y_2](t)} = -\frac{1}{2}(te^{2t} + e^{-t})$$

$$u_2'(t) = \frac{g(t)y_1(t)}{W[y_1,y_2](t)} = \frac{1}{2}(te^{t} + e^{-2t})$$

$$\Rightarrow u_1(t) = -\frac{1}{2}\int te^{2t} + e^{-t}\,dt = -\frac{te^{2t}}{4} + \frac{e^{2t}}{8} + \frac{e^{-t}}{2}$$

$$u_2(t) = \frac{1}{2}\int te^{t} + e^{-2t}\,dt = \frac{te^t}{2} - \frac{e^t}{2} - \frac{e^{-2t}}{4}$$

$$\Rightarrow \psi(t) = u_1y_1 + u_2y_2$$

$$= \frac{te^{3t}}{4} - \frac{3}{8}e^{3t} + \frac{1}{4} = e^{3t}\left(\frac{t}{4} - \frac{3}{8}\right) + \frac{1}{4}$$

step 3: The general solution is then given by.

$$y(t) = c_1e^t + c_2e^{2t} + e^{3t}\left(\frac{t}{4} - \frac{3}{8}\right) + \frac{1}{4}$$

$$y(0) = 1 \Rightarrow c_1 + c_2 - \frac{1}{8} = 1 \Rightarrow c_1 + c_2 = \frac{9}{8}$$

$$y'(0) = \frac{5}{8} \Rightarrow c_1 + 2c_2 - \frac{7}{8} = \frac{5}{8} \Rightarrow c_1 + 2c_2 = \frac{3}{2}$$

$$\Rightarrow c_1 = \frac{3}{4}, \quad c_2 = \frac{3}{8}$$

$$\left(y'(t) = c_1e^t + 2c_2e^{2t} + 3e^{3t}\left(\frac{t}{4} - \frac{3}{8}\right) + \frac{e^{3t}}{4}\right)$$

Thus, $y(t) = \frac{3}{4}e^t + \frac{3}{8}e^{2t} + e^{3t}\left(\frac{t}{4} - \frac{3}{8}\right) + \frac{1}{4}$